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DETAIL SPECIFICATION

COUPLER, OPTICAL

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of an optical coupler used on the M981 Fire Support Team Vehicle (FISTV).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplements thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS

FEDERAL

BB-N-411 - Nitrogen, Technical.

DEPARTMENT OF DEFENSE

MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-PRF-23377 - Primer Coatings: Epoxy, High-Solids.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines (See 4.4.1).

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

11507293 - Rangefinder-Target Designator, Laser MX-9759/TVQ-2 (G/VLLD).
12310400 - Optical Coupler Assembly.
12343606 - ITA Assembly.

(Copies of drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AS1933 - Age Controls for Hose Containing Age-Sensitive Elastomeric Materials (DoD Adopted).

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENT

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Design, materials, and manufacturing. Design, materials, and manufacturing process selection shall be specified herein and in applicable referenced specifications, standards, and drawings. Material shall be free from imperfections or defects which affect their performance and serviceability. The optical coupler shall provide the means for relaying the large ray bundle emerging from the exit pupil of the Ground/Vehicular Laser Locator and Designator (G/VLLD, Drawing 11507293) into the Image Transfer Assembly (ITA, Drawing 12343606) so that an operator can look into the ITA eyepiece and see through the G/VLLD. The input to the coupler in a collimated image ray bundle from the G/VLLD eyepiece. The output of the coupler in an image located at the paraxial image plane within the ITA (see 4.5). The optical coupler shall permit easy dismounting of the G/VLDD from the vehicle without removing the coupler from the ITA.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes the economically advantageous life cycle costs.

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3.2.2 Metals. All metals utilized shall be of corrosion resistant type or suitably protected to resist corrosion during the normal service life under the environmental conditions specified. The use of dissimilar metals which could cause galvanic action shall be protected using the manufacturer's standard process (see 4.5).

3.2.3 Nonmetals. All nonmetals shall be suitable for their intended purpose during the normal life and under the environmental conditions specified. Use of elastomerics shall be governed by the age controls of SAE AS1933 (see 4.5).

3.2.4 Protective window. The optical coupler objective lens shall be covered by a protective window with a protection level equivalent to BK 7 glass or better (see 4.5 and 6.4).

3.2.5 Purge and charge. The optical coupler shall be purged of all the air and the moisture, and charged to an internal pressure of 0 ± 0.2 pounds per square inch gage (psig) (0 ± 1.38 kilopascals (kPa)) with dry nitrogen conforming to BB-N-411, Type I, Class 1, Grade A or B (see 4.5).

3.3 Operating requirements. Each optical coupler shall provide the following functional, operational, and performance capabilities.

3.3.1 Resolution. Resolution specified in 3.3.1.1 and 3.3.1.2 shall be for a high contrast target which shall be the USAF 1951 test chart or the NBS 1010A microcopy test chart (see 4.6.1).

3.3.1.1 Optical coupler resolution. The optical coupler shall clearly resolve an angle of 104 arc seconds or less, on axis (see 4.6.1.1).

3.3.1.2 Integrated system resolution. When the optical coupler is assembled between the G/VLLD and the ITA, the system shall clearly resolve an angle of 8 arc seconds or less, on axis, with the observer's eye anywhere within the exit pupil of the ITA eyepiece. The integrated system shall also clearly resolve the G/VLLD LED data numbers which are approximately in the plane of the G/VLLD reticle (see 4.6.1.2).

3.3.2 Astigmatism. The optical coupler axial astigmatism shall not be greater than 0.2 diopter (see 4.6.2).

3.3.3 Magnification. The optical coupler shall provide a magnification of 1 ±7% from the optical coupler input to the ITA output (see 4.6.3).

3.3.4 Entrance pupil. The optical coupler shall provide an entrance pupil that shall not reduce the integrated system exit pupil to less than 0.22 inch (in.) (5.5 millimeters (mm)) when the integrated system is properly pupil matched (see 4.6.4).

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3.3.5 Field of view. The integrated system, when incorporating the optical coupler, shall provide a real field of view that is not less than 5 degrees (0.087 radians) in any pupil matched condition (see 4.6.5).

3.3.6 Lateral color. The optical coupler shall provide that the lateral color of the system shall not be greater than 3 minutes of an arc out of the system for wavelengths between 0.0000192 in. (486 nanometers (nm)) and 0.0000258 in. (656 nm) (see 4.6.6).

3.3.7 Visual anomalies. The optical coupler shall contribute no visually perceptible defects to the integrated system such as flare ghost image or effects outside the ITA exit pupil (see 4.6.7).

3.3.8 Leak resistance. The helium leak rate for the optical coupler assembly, when mounted to the ITA, or a fixture simulating the ITA, shall be no greater than 0.000025 atmosphere-cubic centimeter (cm^3) per second when tested with an internal atmosphere which is 10 $\pm$1 percent (%) helium with an ITA (or fixture) internal pressure of 1 atmosphere (atm) (101.325 kPa) and an external pressure which is no greater than 1/1500 atm (67.55 Pascals (Pa)) (see 4.6.8).

3.4 Interface requirements. Each optical coupler shall accommodate the following inputs and interfaces.

3.4.1 Configuration. The external configuration of the optical coupler shall not exceed the external configuration shown on Drawing 12310400 without obtaining approval from the contracting authority (see 4.7.1).

3.4.2 Weight. The weight of the optical coupler shall not exceed 9.75 pounds (lb) (4.43 kilograms (kg)) (see 4.7.2).

3.5 Support and ownership requirements. Each optical coupler shall meet the following logistical requirements and possess the following life cycle characteristics.

3.5.1 Paint. All exterior painted surfaces shall be painted with chemical agent resistant aliphatic polyurethane coating conforming to MIL-C-46168, Color No. 383 green, or an equivalent. All aluminum surfaces shall be cleaned and treated per MIL-C-5541, or an equivalent, and primed per MIL-PRF-23377, Type I, or an equivalent (see 4.8.1).

3.5.2 Identification marking. Identification marking shall be permanent and legible and as a minimum, shall include the marking identification code (cage), the contract number, and National Stock Number (NSN). The nameplate shall be permanently attached, in the proper location, and include the following information (see 4.8.2):

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a. Optical coupler
b. Manufacturer's identification
c. Manufacturer's serial number
d. Military part number
e. National stock number (NSN)
f. Manufacturer's part number
g. Date of manufacture
h. Contract number
i. U.S.

3.5.3 Safety. The optical coupler shall pose no physical hazards to personnel operating the assembly. The optical coupler shall be free of structural and welding defects, chips, burrs, sharp edges, process residue, or any other form of contamination that may be harmful to the operator (see 4.8.3)

3.6 Operating environment conditions. The optical coupler shall function as specified when exposed to the environmental conditions specified herein. The optical coupler shall meet the operating requirements of 3.3.1 through 3.3.8 during exposure to those environments designated as operating, as well as, those designated as nonoperating or storage.

3.6.1 Temperature. The optical coupler shall operate in ambient air temperature of +145 degrees Fahrenheit (°F) (63 degrees Celsius (°C)) to -25°F (-32°C). The optical coupler shall be capable of being stored at temperatures of +160°F (71°C) to -40°F (-40°C) (see 4.9.1).

3.6.2 Temperature shock. The optical coupler shall not show any operational degradation after being exposed to temperature shock in accordance with temperature extremes of -40°F (-40°C) to +160°F (71°C) (see 4.9.2).

3.6.3 Humidity. The optical coupler shall operate in humidity conditions of 95% relative humidity (rh) at temperatures from -25°F (-32°C) to +145°F (63°C); and at humidity conditions of 5% rh at temperatures from +70°F (21°C) to +145°F (63°C) (see 4.9.3).

3.6.4 Ambient pressure. The optical coupler shall operate at ambient air pressure of 14.7 to 10.1 pounds per square inch absolute (psia) (100 to 69.6 kPa) which corresponds to altitudes of 0 to 10 000 feet (ft) (0 to 3.05 kilometers (km)). Nonoperating, the optical coupler shall withstand exposure to air pressures from 14.7 to 2.47 psia (100 to 17 kPa) which corresponds to altitudes of 0 to 40 000 ft (0 to 12.2 km) (see 4.9.4).

3.6.5 Rain (nonoperating). The optical coupler shall operate without degradation of performance after exposure to a 12 hour, 9.5 in. (24.1 centimeters (cm)) rainfall which includes intermittent winds of 35 knots (65 kilometers per hour (km/h)) and the following intensities (see 4.9.5):

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Duration	Rate of rainfall
1 minute	27 inches per hour (in./h) (68.6 centimeters per hour (cm/h))
5 minutes	12 in./h (30.5 cm/h)
10 minutes	9 in./h (23 cm/h)
1 hour	5.5 in./h (14 cm/h)

3.6.6 Sand and dust. The optical coupler shall be operational while exposed to wind blown particles 0.000394 to 0.0394 in. (0.01 to 1 mm) in diameter with predominant diameters between 0.00591 and 0.01181 in. (0.15 and 0.30 mm) with winds greater than 30 knots (56 km/h) (see 4.9.6).

3.6.7 Fungus (nonoperating). The optical coupler shall be operational after being exposed to a fungus inoculation for 90 days at a relative humidity of 95% at 84°F (29°C) (see 4.9.7).

3.6.8 Vibration (nonoperating). The optical coupler shall not be damaged nor experience performance degradation after being exposed to vibration levels as shown in Figure 3, with sinusoidal cycling time of 3 hours per axis in accordance with the following schedule. The sweep time shall be 15 minutes from 5 to 500 to 5 Hz (see 4.9.8).

Frequency	Vibration level
5 to 5.5 Hertz (Hz)	1 inch double amplitude
5.5 to 30 Hz	1.5 gravity (g)
30 to 50 Hz	0.033 inch double amplitude
50 to 500 Hz	4.2 g

3.6.9 Shock (nonoperating). The optical coupler shall not be damaged nor experience performance degradation after exposure to severe shock levels with an 11 millisecond (ms) sawtooth pulse of 40 g (peak) shock applied (see 4.9.9).

4. VERIFICATION

4.1 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (CI) (see 4.3).
    1. Examination (see 4.3.2).
    2. Acceptance tests (see 4.3.3).

4.2 First article inspection. The first article inspections shall be performed on preproduction or initial production samples as specified herein (see 3.1). Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply optical coupler assemblies that are fully representative of those inspected as a first article sample.

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Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.2.1 In-process inspection. During fabrication of the first production optical coupler, in-process inspections will be conducted by Government representatives to evaluate conformance of materials and workmanship to requirements of specified documents. The coupler and all components shall be available for inspection. These inspections shall be made at the contractor or subcontractor's facilities prior to application of primer and paint. Processing and welding procedures, quality system, inspection records, calibration procedures, radiographic procedures and welder certifications will be reviewed and evaluated during the in-process inspections.

4.2.2 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two optical coupler assemblies. Preproduction inspection shall consist of inspection as specified in table I.

4.2.3 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two from the first ten optical coupler assemblies produced under the production contract for initial production inspection. Initial production units shall be inspected as specified in table I.

4.3 Conformance inspection. Conformance inspection shall include the examinations 4.3.2 and the tests of 4.3.3.

4.3.1 Sampling. Sampling shall be as specified in the contract or order (see 6.2).

4.3.2 Examination. The samples selected in accordance with 4.3.1 shall be subjected to the examinations specified in table I. Failure to any examination shall be cause for rejection.

4.3.3 Acceptance tests (100% inspection). Each optical coupler shall be subjected to the tests specified in table I. Failure to any test shall be cause for rejection.

Table I. Verification methods.

Title	Requirement	Inspection	First article	CI	
				Exam	Acceptance test
Design, materials and manufacturing	3.2 through 3.2.5	4.5	X	X	
Operating requirements	3.3	4.6			
Resolution	3.3.1	4.6.1	X		
Optical coupler resolution	3.3.1.1	4.6.1.1	X		X
Integrated system resolution	3.3.1.2	4.6.1.2			X
Astigmatism	3.3.2	4.6.2	X		X
Magnification	3.3.3	4.6.3	X		
Entrance pupil	3.3.4	4.6.4	X		

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Table I. Verification methods - Continued.

Title	Requirement	Inspection	First article	CI	
				Exam	Acceptance test
Field of view	3.3.5	4.6.5	X		
Lateral color	3.3.6	4.6.6	X		
Visual anomalies	3.3.7	4.6.7			
Leak resistance	3.3.8	4.6.8	X		X
Interface requirements	3.4	4.7			
Configuration	3.4.1	4.7.1	X		
Weight	3.4.2	4.7.2	X	X	
Support and ownership requirements	3.5	4.8			
Paint	3.5.1	4.8.1	X	X	
Identification marking	3.5.2	4.8.2	X	X	
Safety	3.5.3	4.8.3	X	X	
Operating environment requirements	3.6	4.9			
Temperature	3.6.1	4.9.1	X		
Temperature shock	3.6.2	4.9.2	X		
Humidity	3.6.3	4.9.3	X		
Ambient pressure	3.6.4	4.9.4	X		
Rain	3.6.5	4.9.5	X		
Sand and dust	3.6.6	4.9.6	X		
Fungus	3.6.7	4.9.7	X		
Vibration	3.6.8	4.9.8	X		
Shock	3.6.9	4.9.9	X		

4.4 Verification methods. Acceptable verification methods included in this section are visual inspection and measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.4.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures to verify performance. See the contract for alternatives that replace verification methods required by this specification.

4.4.2 Verification conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 73 ± 18°F (23 ± 10°C).
b. Barometric pressure: 28 (+2, -3) in. (725 (+50, -75) mm) of mercury (Hg).
c. Relative humidity: 50 ± 30 percent.

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4.5 Design, materials and construction. To determine conformance to 3.2 through 3.2.5, inspection and material certification records shall be maintained by the contractor providing proof or verification that design, construction, assembly, processing, and materials conform to the requirements specified herein. Examination shall be visual, tactile, or by measurement with Standard Inspection Equipment (SIE). Scratches, digs, bubbles, inclusions, and cement defects on the optical components shall not exceed the criteria specified within Appendix A. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.6 Operating requirements verification.

4.6.1 Resolution verification.

4.6.1.1 Optical coupler resolution verification. To determine conformance to 3.3.1.1, the optical coupler shall be mounted in a test fixture as shown in figure 1. Using the USAF 1951 test chart or the NBS 1010A microcopy test chart, the high contrast target shall be brought into sharp focus. The smallest resolved pattern shall be noted and the angle of resolution in the object plane shall be calculated. An optical magnifier (microscope) may be used to aid in the observation. The coupler shall clearly resolve an angle of 104 arc seconds or less on axis.

4.6.1.2 Integrated system resolution verification. To determine conformance to 3.3.1.2, the optical coupler, G/VLLD, ITA and collimator shall be mounted onto the alignment fixture as shown in Figure 2. The high contrast test target shall be brought into sharp focus at the intersection of the G/VLLD reticle crosshairs, if a G/VLLD is used. An optical magnifier or dioptometer may be used to aid the observation at the ITA eyepiece. The resolved angle shall be 8 arc seconds or less, on axis, with the observer's eye anywhere within the exit pupil of the ITA eyepiece. The G/VLLD, if used, shall be electrically connected and the lens cover shall be closed to prevent firing of the laser. The G/VLLD LED data numbers shall appear as zeros and shall be clearly resolved, if a G/VLLD is used.

NOTE: Wherever the G/VLLD is referenced in a test, an optical test fixture which is equivalent to the G/VLLD may be used in its place.

4.6.2 Astigmatism verification. To determine conformance to 3.3.2, the optical coupler shall be mounted as shown in Figure 1. The image of the astigmatism target shall be brought into sharp focus. When the image is viewed through an optical magnifier (microscope), readings shall be made when both the vertical lines of the astigmatism target are in focus and when the horizontal lines of the astigmatism target are in focus. The astigmatism, in diopters, is the difference between the two readings. A difference of 0.01 in. (0.254 mm) corresponds to 0.2 diopter. The astigmatism shall not be greater than 0.2 diopter.

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4.6.3 Magnification verification. To determine conformance to 3.3.3, install the optical coupler on the ITA. A telescope coupled to the ITA eyepiece shall be used to bring the target image into sharp focus at the center of the field. The angular size of the target image shall be measured using the calibrated reticle of the telescope. The telescope shall then be repositioned to focus directly on the target and the angular size of the target shall be measured. The magnification shall be calculated as the ratio of the first to the second angular measurements taken. The magnification shall be 1 $\pm$ 7%.

4.6.4 Entrance pupil verification. To determine conformance to 3.3.4, the integrated system configuration shall be used. With the G/VLLD pupil matched to the optical coupler, a uniformly illuminated target, subtending an angle larger than the field-of-view of the G/VLLD, shall be positioned in front of the G/VLLD. An optical magnifier shall be placed behind the eyepiece of the ITA so the exit pupil is brought into sharp focus in the plane of the reticle of the magnifier. Measure and record the diameter of the exit pupil by using the reticle in the magnifier. The G/VLLD shall be removed and the uniformly illuminated target shall be positioned in front of the optical coupler so it covers the field of view of the coupler. Measure and record the diameter of the ITA exit pupil. Reposition the illuminated target in front of the night sight objective lens of the ITA and switch the ITA to the night position. Again measure and record the diameter of the ITA exit pupil. The three exit pupil measurements shall not be less than 0.22 in. (5.5 mm).

4.6.5 Field of view verification. To determine conformance to 3.3.5, the integrated system configuration shall be used. The target image shall be placed at one edge of the G/VLLD optical field and then at the opposite edge. The angular distance thus traversed shall be measured. The field of view shall not be less than 5 degrees (0.087 radians) in any pupil matched condition.

4.6.6 Lateral color verification. To determine conformance to 3.3.6, the integrated system configuration shall be used. A monochromatic target 0.0000192 in. or 0.0000258 in. (486 nm or 656 nm) shall be positioned in front of the G/VLLD and off-axis more than half the angular distance from the central optical axis to the edge of the field. A 5 nm diameter aperture shall then be positioned at the ITA exit pupil. An alignment telescope shall then be positioned behind the aperture, so the target image is located at the intersection of the telescope reticle crosshairs. The F and C filters 0.0000192 in. and 0.0000258 in. (486 nm and 656 nm) shall be alternately positioned to provide alternate monochromatic illumination of the target. The apparent angular displacement of the target image which is caused by the alternate illumination shall be measured with the telescope. The lateral color of the integrated system shall be no greater than 3 minutes of an arc for wavelengths between 0.0000192 in. and 0.0000258 in. (486 nm and 656 nm).

4.6.7 Visual anomalies verification. To determine conformance to 3.3.7, the integrated system configuration shall be used. A 100 watt incandescent lamp shall be positioned at various angular locations outside the G/VLLD field of view but within the G/VLLD forward hemisphere at a distance of not more than 2 ft (610 mm). The ITA eyepiece apparent field shall be visually examined for evidence of flare or ghost images. The positioning of the lamp at various angular

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locations shall be repeated while the ITA eyepiece is observed for evidence of light emerging from outside the exit pupil. No visually perceptible defects shall be observed.

4.6.8 Leak resistance verification. To determine conformance to 3.3.8, the optical coupler shall be placed in a suitable chamber wherein the helium leak rate from the coupler shall be measured. The internal pressure of the coupler shall be 1 atm (101.325 kPa) which includes 10 ± 1% helium. The chamber shall be pumped to vacuum so the external pressure on the coupler shall be no greater than 1/1500 atm (67.55 Pa). The helium leak rate shall be measured and shall not be greater than 0.000025 atm-cubic centimeters per second.

4.7 Interface requirements verification.

4.7.1 Configuration verification. The optical coupler shall be inspected for conformance to Drawing 12310400 using one of the methods outlined in 4.4 or 4.4.1.

4.7.2 Weight verification. To determine conformance to 3.4.2, the optical coupler shall be weighed and shall be no heavier than 9.75 lb (4.43 kg). Weight over the specified value shall be cause for rejection.

4.8 Support and Ownership requirements verification.

4.8.1 Paint verification. One or more of the methods outlined in 4.4 or 4.4.1 shall be used to inspect the application of CARC paint and the color of the optical coupler to conform with the requirement of 3.5.1.

4.8.2 Identification marking verification. To determine conformance to 3.5.2, the optical coupler shall verify the presence of the required markings on the assembly. After performing all operating environmental tests in 4.9 re-inspect the assembly markings for readibility and presence.

4.8.3 Safety verification. To determine conformance to 3.5.3, the optical coupler shall be inspected for hazardous burrs, chips, nicks, sharp edges, foreign materials, or other imperfections that pose physical danger to the operator. Inspect the coupler for exposed, frayed, unsecured or improperly fitted fasteners.

4.9 Operating environment requirement verification.

4.9.1 Temperature verification. To determine conformance to 3.6.1, the optical coupler shall be tested as follows:

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a. Operation:
   1. Low temperature. The optical coupler shall be subjected to and pass the test in accordance with MIL-STD-810, Method 502.3, Procedure II (see 4.4.1) at -25°F (-32°C).
   2. High temperature. The optical coupler shall be subjected to and pass the test in accordance with MIL-STD-810, Method 501.3, Procedure II (see 4.4.1) at 145°F (63°C).

b. Storage:
   1. Low temperature. The optical coupler shall be subjected to and pass the test in accordance with MIL-STD-810, Method 502.3, Procedure I (see 4.4.1) at -40°F (-40°C) for 6 hours.
   2. High temperature. The optical coupler shall be subjected to and pass the test in accordance with MIL-STD-810, Method 501.3, Procedure I (see 4.4.1) at 160°F (71°C) for 4 hours.

4.9.2 Temperature shock verification. To determine conformance to 3.6.2, the optical coupler shall be subjected to and pass the test in accordance with MIL-STD-810, Method 503.3, Procedure I (see 4.4.1), except that the low temperature shall be -40°F (-40°C). To verify operational integrity, the resolution test (see 4.6.1), astigmatism test (see 4.6.2) and pressurization test (see 4.6.8) shall be performed after this test.

4.9.3 Humidity verification. To determine conformance to 3.6.3, the optical coupler shall be operated successfully during the humidity test in accordance with MIL-STD-810, Method 507.3, Procedure II (see 4.4.1), with the exception that the high temperature shall be 145°F (63°C) and the low temperature shall be -25°F (-32°C) and number of test cycles shall be ten. The test shall be repeated with a low relative humidity of 5% at temperatures of +70°F (21°C) and +145°F (63°C).

4.9.4 Ambient pressure verification (altitude). To determine conformance to 3.6.4, the optical coupler shall be subjected to and pass the test in accordance with MIL-STD-810, Method 500.3 (see 4.4.1), with the following:

a. Procedure I. The air pressure shall be equivalent to an altitude of 40 000 ft (12.2 km).
b. Procedure II. The air pressure shall be equivalent to an altitude of 10 000 ft (3.1 km).

4.9.5 Rain verification (nonoperating). To verify conformance to 3.6.5, the optical coupler shall be sujected to and pass a rain test in accordance with MIL-STD-810, Method 506.3, Procedure I (see 4.4.1), with the exception that the test conditions shall be as specified in 3.6.5.

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4.9.6 Sand and dust verification. To determine conformance to 3.6.6, the optical coupler shall be operated successfully during a sand and dust test in accordance with MIL-STD-810, Method 510.3, Procedure I (see 4.4.1), with the exception that sand and dust particle size shall be as specified in 3.6.6, with a density of 0.000000006 grams per cubic centimeter (g/cm^3) and the temperature in test step 5 shall be 145°F (63°C).

4.9.7 Fungus verification (nonoperating). To verify conformance to 3.6.7, the optical coupler shall be subjected to and pass the fungus test at 84°F and 95% rh with fungi as specified in MIL-STD-810, Method 508.4, for a period of 90 days. No test shall be required if the manufacturer can certify that only fungus inert materials have been used; or if the manufacturer can present certification of anti-fungal treatment acceptable to the Government (see 4.4.1 and 6.3).

4.9.8 Vibration verification (nonoperating). To verify conformance to 3.6.8, the optical coupler shall be subjected to and pass the test in accordance with MIL-STD-810, Method 514.4, Procedure I, Category 8 (see 4.4.1), with the test level as specified in Figure 3 with a sinusoidal cycling time of 3 hours per axis. Sweep time shall be 15 minutes from 5 to 500 to 5 Hz. Vibration amplitudes shall be as specified in 3.6.8.

4.9.9 Shock verification (nonoperating). To verify conformance 3.6.9, the optical coupler shall be subjected to and pass the tests in accordance with MIL-STD-810, Method 516.4, Procedure I (see 4.4.1). The 11 ms sawtooth pulse with 40 g peak of Figure 516.4-4 shall be used.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

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6.1 Intended use. The optical coupler, Drawing 12310400, is intended for use on the M981 Fire Support Team Vehicle (FISTV) to interface between the G/VLLD and ITA. The optical coupler is military unique because it is an integral part of a system that is designed strictly for military purposes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of Department of Defense Index of Specifications and Standards (DoDISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, and 2.3).
c. Part number of the optical coupler required (see 2.2.2 and 3.2)
d. When first article is required (see 3.1).
e. If preproduction inspection is required (see 4.2.2).
f. If initial production inspection is not required (see 4.2.3).
g. If inspection conditions are other than as specified (see 4.4.2).
h. Packaging requirements (see 5.1).

6.3 Definition. For the purpose of this specification, the following definition will apply:

a. BK 7. One of the most common and widely used borosilicate crown glasses available. It performs well against most chemical tests, it is relatively hard, and does not scratch easily. BK 7 glass can be handled without the need for special precautions. The bubble and inclusion content of BK 7 is extremely low.
b. Fungus inert material. A material which, in all modified stated and grades, is not a nutrient to fungi.

6.4 Subject term (key word) listing.

Cover
Lenses
Mirror
Prism
Seal
Visual anomalies
Window

6.5 Changes from previous issue. Marginal notations are not used in this version to identify changes with respect to the previous issue due to the extent of the changes.

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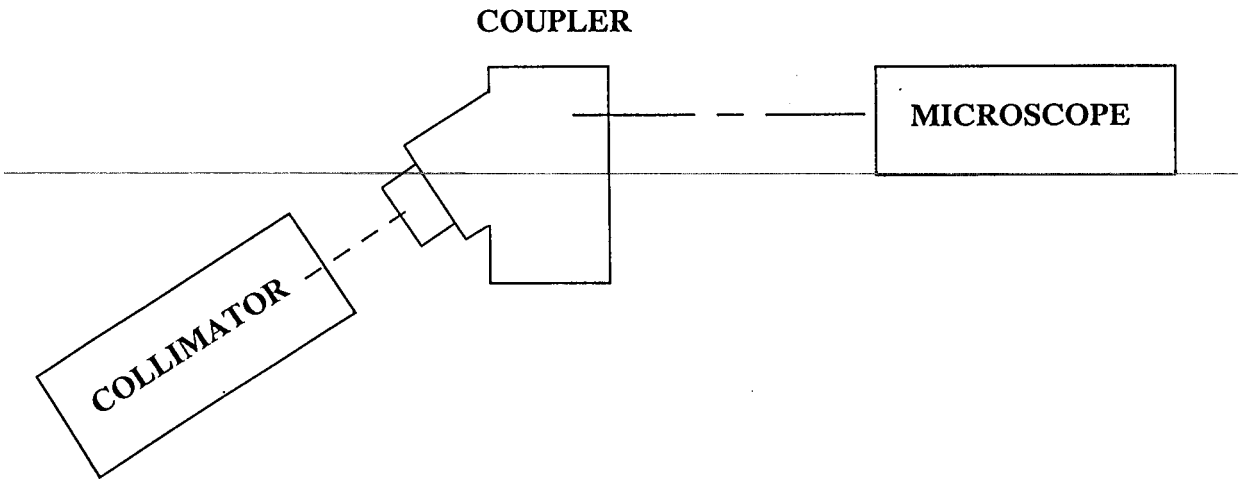


FIGURE 1. Optical coupler test configuration.

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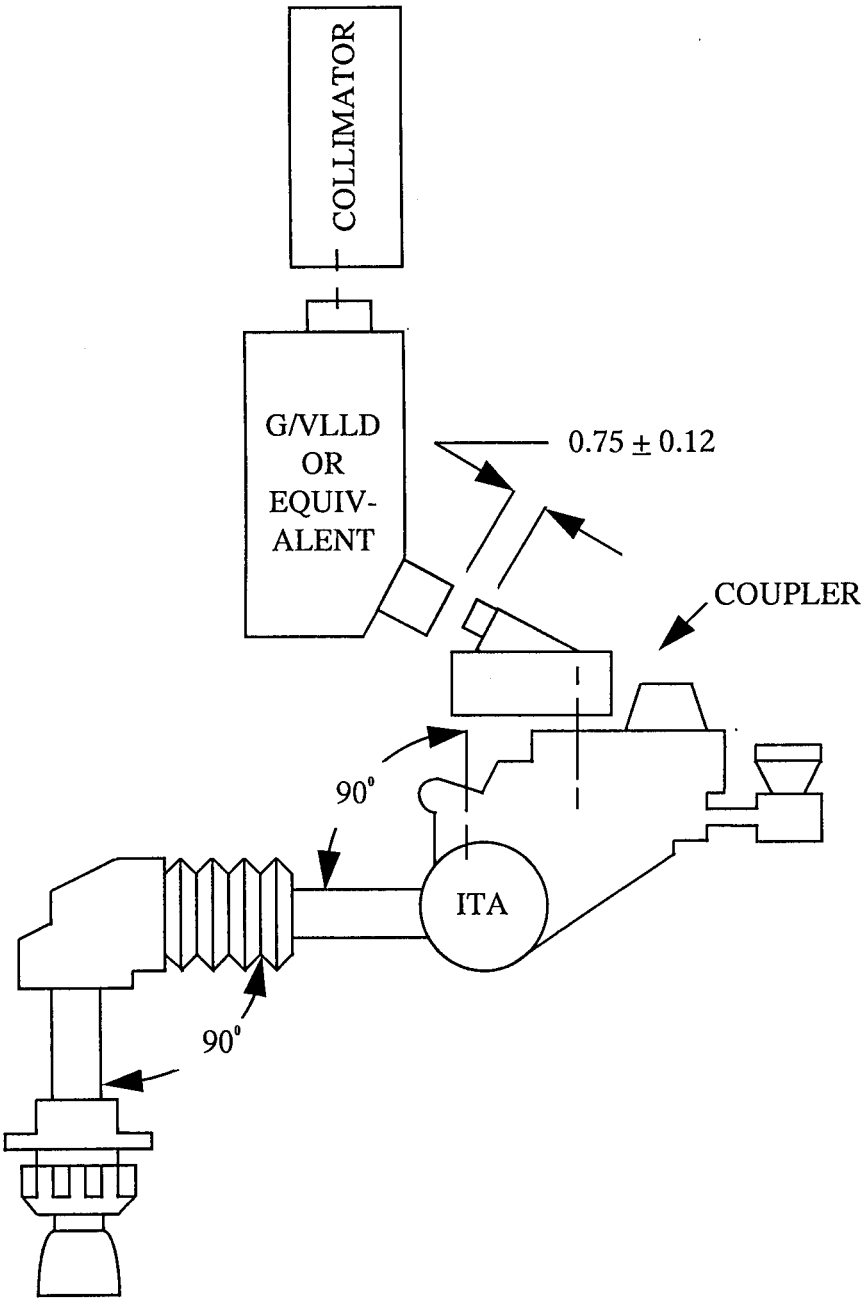
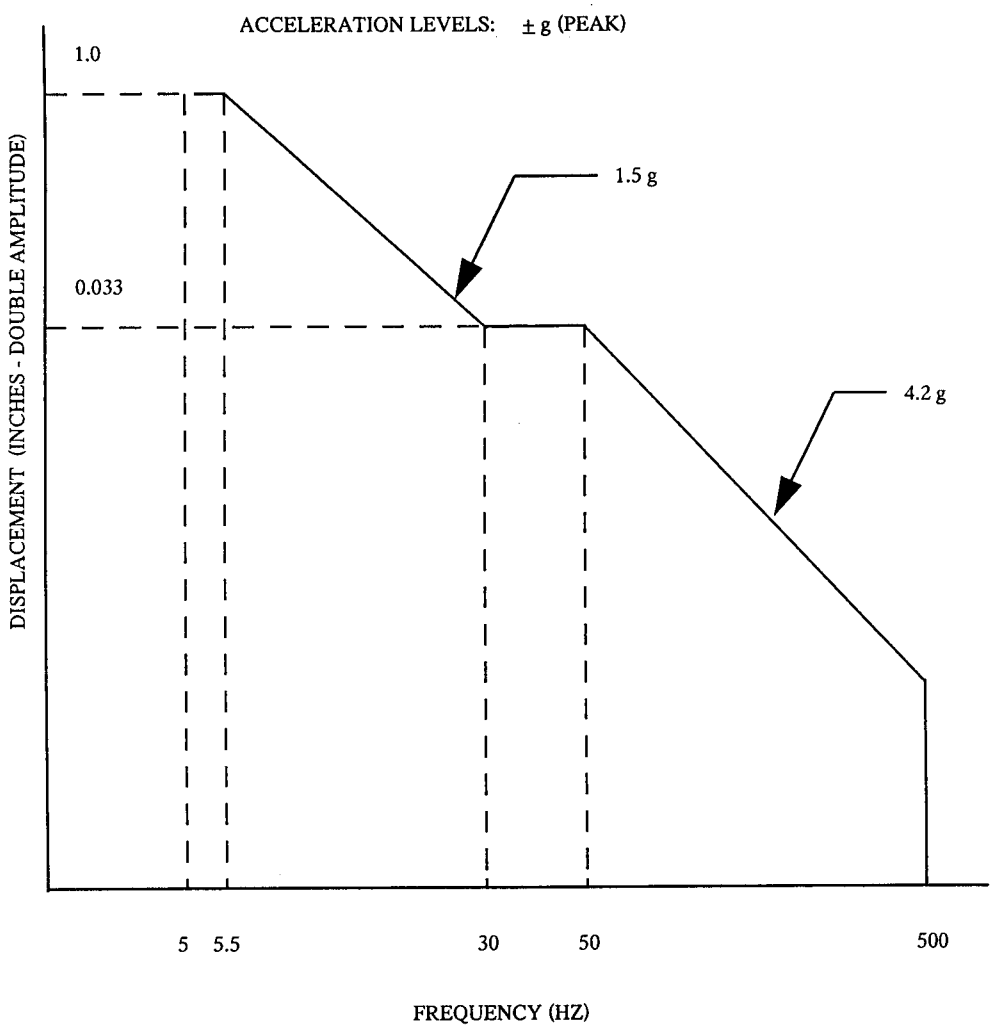


FIGURE 2. Integrated system test configuration.

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NOTE: All curves shall be extended to 2 Hz when item resonances below 5 Hz are expected.

FIGURE 3. Vibration test curve.

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APPENDIX A

CRITERIA TO DETERMINE SURFACE QUALITY OF OPTICAL LENSES, PRISMS, MIRRORS, RETICLES, AND WINDOWS

A.1 SCOPE

A.1.1 Scope. The appendix details the criteria to determine surface quality of the optical components of the optical coupler. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

A.3 CONDITIONS

A.3.1 Optical drawings and diagrams. Component optical drawings shall indicate surface quality, and optical system diagrams shall indicate the diameter of an axial beam of rays.

A.3.2 Designation of defect size. Limiting sizes of surface defects shall be designated on the drawings by two numbers which refer to two graded sets of surface quality standards. The first number shall refer to the scratches and the second number shall refer to digs.

A.3.3 Scratches.

A.3.3.1 Circular element. The combined length of maximum size scratches located on each surface of an optical element shall not exceed one quarter the diameter of that element.

A.3.3.1.1 Maximum combined lengths of scratches. When a maximum size scratch is present, the sum of the products of the scratch numbers times the ratio of their length to the diameter of the element or appropriate zone shall not exceed one half the maximum scratch number. When a maximum size scratch is not present, the sum of the products of the scratch numbers times the ratio of their length to the diameter of the element or appropriate zone shall not exceed the maximum scratch number.

A.3.3.2 Noncircular shaped element. The computing diameter of element shapes other than circular shall be that of a circle or equal area. Scratches beyond the free aperture of any element as given on the optical system drawings or detail drawings shall not be considered when applying the appropriate formula specified in A.3.3.1.1.

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A.3.3.2.1 True roof surfaces on prisms. True roof surfaces on prisms shall be considered equivalent to a single surface equal to the sum of the individual roof areas for purposes of the scratch and dig computation, except that the roof edge shall not be considered in the summation of the length of the allowable scratches. Scratch and dig tolerances for roof prisms are set on the basis that the equivalent surface above is viewed from the air side.

A.3.3.2.2 Surface quality, central zone. Areas of surfaces whose specified scratch qualities are 20 or better shall have no more than 4 separate scratches in any 1/4-in. diameter circular area. This requirement does not apply for scratches smaller than number 10.

A.3.3.3 Surface quality, outer zone. Surface quality outside the free aperture of any element shall be considered 80-50, unless otherwise required.

A.3.3.4 Coating scratches. Coating scratches, scratches which do not penetrate the glass surface, shall be within the same limits specified in A.3.3. Coating scratches shall be considered separate from the substrate scratch requirements.

A.3.4 Digs.

A.3.4.1 Dig designation. Dig number are the actual diameters of defects allowed, specified in units of 1/100 mm. In the case of irregular shaped digs, the diameter shall be taken as the average of the maximum length and minimum width.

A.3.4.2 Maximum size digs. The permissible number of maximum size digs shall be 1 per each 20 mm of diameter or fraction thereof on any single optical surface. The sum of the diameters of all digs as estimated by the inspector shall not exceed twice the diameter of the maximum size specified per 20 mm diameter. Digs less than 2.5 microns shall be ignored.

A.3.4.3 Surface quality. All digs on each surface whose dig quality is number 10 or smaller shall be separated edge to edge by at least 1 mm. The measurement of scattering shall not be required for surfaces where digs larger than number 10 are allowed.

A.3.5 Bubbles and inclusions. Bubbles shall be classified as surface digs. Any inclusion in the glass shall be treated as a bubble. The size of irregular shaped inclusions shall be considered as 1/2 the sum of the maximum length and the maximum width. Bubble size tolerances are identical in all respect to digs; but the bubble tolerance, is an addition to the dig tolerance.

A.3.5.1 Maximum size bubbles. The permissible number of maximum size bubbles shall be 1 per 20 mm of light path, or fraction thereof, of any single element. The sum of the diameters of all bubbles as estimated by the inspector shall not exceed twice the diameter of the maximum size

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specified per 20 mm diameter for each 20 mm of light path. When surface dig quality is 10 or smaller, bubbles shall follow requirements for digs as specified in A.3.4.3.

A.3.6 Cement defects. Cement bubbles, voids, undissolved particles, dry spots, blisters, dirt (lint or dust) within the free aperture of the cemented lens shall not exceed the limits of the defects for digs and bubbles specified in A.3.4.1 thru A.3.5.1 inclusive.

A.3.6.1 Surface quality of cemented face. Cement defects inside the free aperture shall be considered in the basis that the cement interface is a single surface of the specified surface quality. When not specified, the surface quality for a cement face shall be intermediate between that of adjacent faces.

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2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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COUPLER, OPTICAL

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

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c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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